

Rahul Moka

rahulmoka.com | rahulbabu.moka@gmail.com | linkedin.com/in/rahul | [Github/Rahulbruh](https://github.com/Rahulbruh) | [LeetCode/rmoka123](https://leetcode.com/rmoka123)

EDUCATION

Permanent Resident

Bachelors in Computer Science

Memphis, TN

University of Memphis

Aug. 2023 – May 2027

– Dean's List

– **Related Coursework:** **COMP 4720** - Intro to Artificial Intelligence, **COMP 4030** - Design/Analysis of Algorithms, **COMP 3410** - Computer Organization and Architecture, **COMP 2700** - Discrete Structures

EXPERIENCE

Software Engineer, Fullstack Intern – Triad Tech

May-Jul 2026

Reference: matt@triad.tech, linkedin

- Developed and integrated an internal project organization system using **Ruby on Rails** and **React**, managing **7 projects** for **4 clients** and centralizing information on the 'Cork-Board' platform.
- Automated internal workflows using **MCP server**, leading to an estimated **30%** increase in team efficiency for project management and client interactions.
- Engineered a **Raspberry Pi 5** to send requests to an Oat Foundry Board via **Split-Flap API** in order to create a faster method of displaying on the board for the client, containerized the code via **Docker** to run on boot and connected **Claude** via **SSH** in order to make code edits with agents remotely

Software Engineer Intern – West TN Consulting

Jan. 2025 – Present

Reference: miclaineffeler@westtn.consulting, linkedin

- Automated contract generation for **30+ clients** via a **Next.js**, **React**, and **Node.js** pipeline, saving 30 minutes per contract by parsing form state into server-side templates and PDFs.
- Refactored and improved a complex React code-base by fixing **20 critical bugs** and addressing performance issues in a challenging, unorganized project.

PROJECTS

Self-Maintaining Stripe API Pipeline | *Multi-Agent Automated Migration System Repo*

2026

- Architected a **5-agent pipeline** (Planning, Retriever, Synthesizer, Critic/Verifier, Orchestrator) that detects **Stripe API** breaking changes and autonomously generates fix PRs, using the **Anthropic Claude API**, **ChromaDB (vector DB)** for semantic retrieval, **Python AST** analysis, and **Docker + GitHub CLI** for sandboxed testing and PR delivery.
- Validated against a synthetic repo modeling a real Stripe deprecation (2020 Plans→Prices): reached **100% precision** on usage detection and verified fixes via sandboxed **pytest**, correctly distinguishing stale-test failures from real regressions before opening a human-review-gated PR.

SDXL Cyberpunk-Style Image-Gen LoRA Fine-Tune | *generate here | sdxl-repo*

Jul. 2026

- Fine-tuned **Stable Diffusion XL** via **LoRA (Diffusers, PEFT)** to specialize image generation toward a custom visual style, training a low-rank adapter on the UNet rather than the full **2.6B+ parameter model**
- Engineered a memory-constrained training pipeline in **PyTorch** and **Accelerate** (bf16 mixed precision, gradient checkpointing, 8-bit Adam via **bitsandbytes**, gradient accumulation) trained using **24GB GPU** via a pod from **runpod**
- Built the end-to-end pipeline from scratch: dataset curation/preprocessing with **Hugging Face Datasets**, checkpoint-based overfitting analysis across training runs, and a CLI plus **Gradio** web app for inference

Remote Dev & Personal Automation System (Raspberry Pi) | *Rahul/wol-trigger*

Jul. 2026

- Built a **C++** remote development system using **Boost.Asio** and **PTY/ConPTY** to expose a live terminal session on a home machine, controllable from anywhere via a **Raspberry Pi** acting as an always-on gateway.
- Configured **Wake-on-LAN** and an **iOS Shortcut** to remotely power on the home PC from sleep, then connected via **Termius** over **SSH** to establish a terminal session directly from an iPhone, including launching **Claude Code** to work on projects remotely.
- Integrated the **Apple Watch (HealthKit) API** to continuously log heart rate and calories burned, and the **Canvas LMS API** to automatically verify assignment submission status and retrieve turned-in files.
- Integrated **Claude API** to programmatically compare submitted assignment files against instructor-provided directions, flagging missing or incomplete requirements before grading.

TECHNICAL SKILLS

Programming Languages: C++, Ruby, Java, Python, JavaScript, TypeScript, HTML, CSS

Libraries & Frameworks: PyTorch, Hugging Face, React.js, Next.js, Node.js, Django, Flask, Ruby on Rails, Pandas, NumPy, Pygame, Celery

Cloud & DevOps: AWS, Docker, GitHub CLI, SSH

AI/ML & Agentic Tooling: Machine Learning, Claude API, MCP (Model Context Protocol), ChromaDB (Vector DB), Hugging Face (Datasets, Diffusers, PEFT), Accelerate, Gradio, pytest

Developer Tools: Git, VS Code, Cursor, GitHub Copilot

Core CS: Data Structures & Algorithms, System Design